

🌟 Section: Our Dynamic Universe

💎 Topic: Gravitation and Orbital Motion

Given:

- Radius of orbit:

$r = 7.00 \times 10^7 \text{ m}$

- Gravitational field strength at that altitude:

$g = 4.10 \text{ N kg}^{-1}$

- Gravitational acceleration at Earth's surface:

$g_0 = 9.8 \text{ N kg}^{-1}$

- Radius of Earth:

$R = 6.4 \times 10^6 \text{ m}$

(a) Show that the mass of Earth is $6.10 \times 10^{24} \text{ kg}$

✔ 3 marks

Use the gravitational field strength formula:

$g = \frac{GM}{r^2} \Rightarrow M = \frac{gr^2}{G}$

Substitute:

- $g = 4.10 \text{ N kg}^{-1}$
- $r = 7.00 \times 10^7 \text{ m}$
- $G = 6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$

$$\begin{aligned} M &= \frac{4.10 \times (7.00 \times 10^7)^2}{6.67 \times 10^{-11}} \\ &= \frac{4.10 \times 4.90 \times 10^{15}}{6.67 \times 10^{-11}} \\ &= \frac{2.009 \times 10^{16}}{6.67 \times 10^{-11}} \\ &= 6.10 \times 10^{24} \text{ kg} \end{aligned}$$

✔✔ Matches the expected value — 3 marks awarded

(b) Show that the mass of the satellite is $1.40 \times 10^3 \text{ kg}$

✔ 2 marks

We know:

- $W = mg$
- $g = 4.10 \text{ N/kg,}$
- $W = 5740 \text{ N}$

$m = \frac{W}{g} = \frac{5740}{4.10} = 1.40 \times 10^3 \text{ kg}$

✔✔ Confirmed

(c) Calculate the kinetic energy of the satellite

✔ 3 marks

$E_k = \frac{1}{2}mv^2$

First, calculate orbital speed:

$v = \sqrt{gr} = \sqrt{4.10 \times 7.00 \times 10^7} = \sqrt{2.87 \times 10^8} = 1.69 \times 10^4 \text{ m/s}$

Now:

$$\begin{aligned} E_k &= \frac{1}{2} \cdot 1.40 \times 10^3 \cdot (1.69 \times 10^4)^2 \\ &= 0.5 \cdot 1.40 \times 10^3 \cdot 2.86 \times 10^8 \\ &= 2.00 \times 10^{11} \text{ J} \end{aligned}$$

🧠 Revision Tips

- Use $g = \frac{GM}{r^2}$ when working with gravitational fields above Earth's surface
- Orbital speed shortcut: $v = \sqrt{gr}$ (very useful!)
- Kinetic energy in orbit: $E_k = \frac{1}{2}mv^2$, where v is orbital speed